



ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *Lyme*, Ultimate Guide

A Complete Functional Medicine Guide to Accurate Testing, Co-Infection Assessment, Drainage-First Recovery, and the Lyme-Thyroid Connection

Dr. Adam Seay, DC

Seay Wellness / Root Health, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

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The Test Said Negative. The Body Tells a Different Story.

Lyme disease is one of the most contested diagnoses in modern medicine, and the controversy is not primarily scientific. It is institutional. The science on *Borrelia burgdorferi*, on tick-borne co-infections, and on the limitations of the standard two-tier Lyme test is substantially more settled than the public clinical conversation suggests. The patients caught in the middle of this controversy are those who present with a constellation of symptoms that fits no other diagnosis, who have been exposed to tick-endemic environments, whose standard tests are negative, and who are told by successive providers that their symptoms are unexplained, stress-related, or psychiatric.

The standard Western Blot test was designed with epidemiological specificity in mind: to minimize false positives in the general population during acute illness. For this purpose, it performs adequately. For chronic or late-disseminated Lyme disease, it fails a clinically significant percentage of real cases. The test analyzes a limited panel of *Borrelia* antigens. It tests nothing for the co-infections that are transmitted by the same ticks and that are present in a substantial proportion of chronic Lyme patients. And it was standardized using serum from patients in the acute stage of illness, a stage at which antibody profiles differ substantially from those of patients with late or persistent infection.

What makes chronic tick-borne illness clinically complex is not primarily the *Borrelia* organism itself. It is the co-infection picture. Babesia, Bartonella, Ehrlichia, and Anaplasma are transmitted by the same Ixodes ticks. Each produces a distinct clinical syndrome. Each requires different treatment approaches. A patient infected with both *Borrelia* and Babesia who is treated only for *Borrelia* will see limited improvement because the organism driving their most disabling symptoms, the profound fatigue, the air hunger, the cyclical flares, is not being addressed. This is the most consistent reason why Lyme treatment fails in clinical practice: incomplete pathogen identification leading to incomplete treatment.

This guide walks through eight keys to understanding and addressing tick-borne illness from the ground up. It covers why the standard test fails, what comprehensive testing actually reveals, how co-infections change the clinical picture, the role of the gut in both Lyme pathophysiology and Lyme recovery, the drainage-first framework that prevents treatment-induced crises, and the often-missed connection between Lyme disease and Hashimoto's thyroiditis that affects a meaningful proportion of chronic Lyme patients. Read it as a framework for understanding your own clinical picture.

Why the Standard Test Fails

The two-tier testing protocol recommended by the Centers for Disease Control for Lyme disease consists of an ELISA screening test followed by a Western Blot confirmation test in patients who screen positive. This protocol was designed in the 1990s for a specific purpose: surveillance and outbreak detection in populations with high background exposure. Its application as a clinical diagnostic tool for individual patients, particularly those with chronic symptoms and possible late-disseminated or persistent infection, has been the source of ongoing clinical controversy precisely because the test's design parameters do not match that clinical use case.

The Sensitivity Problem

The ELISA screening test that begins the two-tier protocol has documented sensitivity of approximately 30 to 40 percent for early Lyme disease.¹ For late-disseminated Lyme, sensitivity estimates in independent research range from 45 to 65 percent depending on the study population and the elapsed time since infection. This means that between one-third and one-half of patients with confirmed late Lyme disease will have a negative ELISA and will never proceed to Western Blot confirmation under the standard two-tier protocol. They receive a negative test result and are sent home.

The Western Blot confirmation test compounds this problem. The CDC surveillance criteria for a positive Western Blot require the presence of 5 of 10 specific immunoglobulin G bands for IgG, or 2 of 3 specific bands for IgM. These criteria were designed to maximize specificity for surveillance purposes. In individual clinical diagnosis, they exclude patients with partial antibody responses, patients whose immune profiles have shifted over time, and patients infected with *Borrelia* strains not fully represented in the original band selection criteria. The standard Western Blot tests 10 *Borrelia* antigens. The Vibrant Tickborne 2.0 PCR panel tests 41.

The Co-Infection Blind Spot

The most significant limitation of the standard Lyme test is not its sensitivity for *Borrelia*. It is what the test does not test for at all. The standard two-tier protocol tests for *Borrelia burgdorferi* and nothing else. It does not test for *Babesia*. It does not test for *Bartonella*. It does not test for *Ehrlichia*. It does not test for *Anaplasma*. These organisms are transmitted by the same Ixodes ticks that transmit *Borrelia*, and they are present in a clinically significant proportion of tick-borne illness patients.

Babesia microti and *Babesia duncani* are intraerythrocytic parasites, organisms that infect red blood cells in a manner analogous to malaria. They produce a clinical syndrome characterized by profound cyclical fatigue, night sweats, air hunger, headaches, and fevers that does not respond to standard Lyme antibiotics because antiprotozoal agents, not antibiotics, are required to treat *Babesia*. A patient with both *Borrelia* and *Babesia*

who is treated only with antibiotics targeted at *Borrelia* will experience partial improvement at best, because the organism most responsible for their fatigue and air hunger is not responsive to the treatment protocol.

Bartonella henselae and *Bartonella quintana* produce a neurological and psychiatric symptom profile that is among the most commonly misattributed in all of medicine. Tingling and burning sensations, anxiety, insomnia, mood instability, brain fog, cognitive dysfunction, and OCD-like intrusive thoughts are classic *Bartonella* presentations that are routinely attributed to anxiety disorders or psychiatric conditions. Patients with active *Bartonella* frequently spend years in mental health treatment before the tick-borne connection is identified. Standard Lyme panels miss *Bartonella* entirely.

Ehrlichia and *Anaplasma* are obligate intracellular bacteria that infect white blood cells. They produce an acute febrile illness that can be severe but is generally responsive to doxycycline.² In chronic presentations, they produce a pattern of persistent immune dysfunction and inflammatory burden that is clinically important to identify because it affects treatment sequencing and duration.

The Co-Infection Picture

Understanding the specific clinical presentations of each major tick-borne co-infection is essential for interpreting symptoms accurately and for understanding why a Lyme treatment protocol that does not address co-infections produces limited results. The following descriptions are not exhaustive but capture the clinical features most relevant to patients with chronic tick-borne illness.

Borrelia Burgdorferi: The Primary Lyme Organism

Borrelia burgdorferi is a spirochete, a corkscrew-shaped bacterium with a remarkable capacity for immune evasion. It can alter its surface proteins to evade antibody recognition, invade intracellular environments where antibiotics penetrate poorly, and form biofilm-protected aggregates that resist standard antibiotic concentrations. These adaptations explain why short-course antibiotic treatment produces lasting resolution in early Lyme but frequently fails to eliminate the organism in late-disseminated infection.

The classic bull's-eye rash, erythema migrans, appears in approximately 70 to 80 percent of early Lyme infections but is frequently not observed because it appears at the site of the tick bite, which may be in an area the patient cannot easily examine. The absence of a rash does not rule out *Borrelia* infection. The rash, when present, may also appear as a non-classic red oval or expanding redness without the central clearing that produces the classic target appearance.

Chronic *Borrelia* infection produces the constellation most associated with Lyme disease in the public understanding: migratory joint pain, neurological symptoms including brain fog and cognitive impairment, fatigue that is disproportionate to activity level, and sleep disturbances. The migratory quality of the joint pain, moving from location to location over days to weeks, is particularly characteristic and helps distinguish it from inflammatory arthritides that tend to affect consistent locations.

Babesia: The Overlooked Parasite

Babesia is arguably the most clinically disabling of the tick-borne co-infections in terms of its effect on daily function. The organism infects red blood cells and reproduces within them, periodically releasing new organisms in cycles that produce the characteristic symptom pattern: cyclical episodes of severe fatigue, drenching night sweats, air hunger, low-grade fever, headaches, and emotional lability. Between cycles, patients may feel relatively functional. During cycles, they may be profoundly debilitated.

Air hunger, the sensation of being unable to take a satisfying breath despite normal oxygen saturation, is highly characteristic of *Babesia* and helps distinguish it from *Borrelia*-dominant presentations. Patients often describe feeling as though they are breathing through a wet blanket or cannot fully fill their lungs. This symptom is often attributed to anxiety. When it co-occurs with cyclical fatigue and night sweats, *Babesia* should be in the differential regardless of standard Lyme test results.

Babesia is treated with antiprotozoal combinations, typically atovaquone and azithromycin, not the doxycycline and amoxicillin regimens used for Borrelia.³ A patient being treated for Borrelia with standard antibiotics while carrying active Babesia will see improvement in some symptoms, those driven by Borrelia, and persistent disability in others, those driven by Babesia, because the organisms require fundamentally different treatment approaches.

Bartonella: The Neuropsychiatric Infection

Bartonella is transmitted both by ticks and by cat scratch (from fleas carrying the organism). In the context of tick-borne illness, Bartonella henselae and Bartonella quintana are the primary species. Bartonella infects endothelial cells lining blood vessels and produces a neurological and psychiatric symptom profile that is unique among the tick-borne co-infections.

The neurological signature of Bartonella includes tingling, burning, or electric sensations in the extremities and across the skin surface that patients often describe as “fizzing” or feeling like they are being mildly electrocuted. Profound anxiety, sometimes approaching panic disorder severity, is extremely common. Intrusive, repetitive, or OCD-like thoughts that emerged or worsened with illness onset are a Bartonella signature that most psychiatrists have not been trained to recognize as infectious in origin. Insomnia, particularly difficulty maintaining sleep, is near-universal in active Bartonella infection.

The clinical significance of recognizing Bartonella is substantial because patients with active Bartonella who are in psychiatric treatment, often for anxiety or OCD, are receiving symptomatic management while the underlying infectious driver continues. The psychiatric symptoms in Bartonella are not functional disorders. They are the neurological consequences of an organism with tropism for central nervous system vasculature. Addressing the infection is required for the psychiatric symptoms to resolve.

Ehrlichia and Anaplasma

Ehrlichia and Anaplasma are obligate intracellular bacteria transmitted by ticks that infect white blood cells. They produce an acute febrile syndrome with headache, myalgia, leukopenia, thrombocytopenia, and elevated liver enzymes. In acute presentation, they are generally recognized and treated with doxycycline. In chronic presentations, they contribute to a persistent state of immune dysfunction characterized by impaired neutrophil function, which reduces the immune system’s capacity to control Borrelia and other co-infections. Identifying and treating Ehrlichia and Anaplasma is important as part of the complete co-infection picture because their effect on immune function affects the clinical response to Borrelia treatment.

The Right Testing Framework

Accurate pathogen identification is the prerequisite for effective treatment sequencing. The organisms present, their relative burden levels, and the co-infection combination determine which antimicrobial agents are appropriate, in what order, and for how long. Testing that provides an incomplete picture produces treatment plans with predictable gaps. The comprehensive testing framework at Root Health is designed to characterize the full tick-borne burden before any antimicrobial intervention begins.

The Vibrant Tickborne 2.0 PCR Panel

The Vibrant Tickborne 2.0 PCR panel represents the most comprehensive commercially available tick-borne testing. It uses polymerase chain reaction technology to detect the DNA of tick-borne organisms directly, rather than relying solely on the antibody response, which can be diminished in chronic infection or in patients with immune dysregulation. The PCR approach detects active infection even when the antibody response is insufficient to produce a positive result on standard serology-based tests.

It is important to understand that PCR detects DNA in circulating blood. Spirochetes and other tick-borne organisms that have migrated into joint fluid, connective tissue, or central nervous system compartments may not be detectable in a blood draw at any given moment. A negative PCR result does not rule out infection when symptom burden is clinically consistent with tick-borne illness. In those cases, the full clinical picture, including symptom pattern, exposure history, and response to empirical support, informs the decision-making alongside the laboratory data.

The panel tests for *Borrelia burgdorferi* across 41 antigens, compared to the 10 on a standard Western Blot. The significantly expanded antigen coverage means that *Borrelia* strains and antibody patterns that would produce a negative or indeterminate result on standard testing may produce a positive result on the Tickborne 2.0 panel. It tests for *Babesia microti* and *Babesia duncani* separately, recognizing that the two species have different geographic distributions and potentially different clinical profiles. It tests for *Bartonella henselae* and *Bartonella quintana*. It covers *Ehrlichia* and *Anaplasma* species, tick-borne encephalitis, and relapsing fever organisms. At \$1,406 retail, it provides the breadth of coverage necessary to build a complete clinical picture rather than a partial one. It is available through rootcauseunlocked.com/order-labs.

Complementary Panels

Beyond the Tickborne 2.0 panel, several complementary assessments provide context that affects treatment decisions. The Total Tox Burden panel from Vibrant Wellness at \$989 characterizes the mycotoxin, heavy metal, and environmental chemical burden that coexists with Lyme disease in a high proportion of cases. Mold toxin exposure and Lyme infection are not coincidentally correlated. Mycotoxins suppress immune function, impairing the immune system's capacity to control *Borrelia* and co-infections. The inflammatory burden of active Lyme also impairs the liver's capacity to process mycotoxins. The two conditions maintain

and amplify each other. Identifying the mold burden determines whether antimold and drainage interventions need to run alongside the antimicrobial protocol.

The Organic Acids Test at \$571 provides the neurochemical picture: neurotransmitter metabolite status, mitochondrial function markers, and fungal overgrowth indicators that collectively characterize how the tick-borne burden has affected the brain's biochemical environment. For patients with significant neurological or psychiatric symptoms, the OAT provides the substrate data that guides neurological support within the broader protocol.

For patients with thyroid symptoms alongside tick-borne illness, the Vibrant Hormone Zoomer at \$710 establishes the full thyroid picture including TPO and thyroglobulin antibodies. The Lyme-Hashimoto's connection is covered in detail in Key 7. The clinical decision to add thyroid testing should be made for any Lyme patient with fatigue, weight changes, brain fog, cold intolerance, hair thinning, or mood instability that has persisted despite Lyme-focused treatment.

Panel	What It Reveals	Price
Vibrant Tickborne 2.0 PCR	Borrelia (41 antigens), Babesia microti and duncani, Bartonella, Ehrlichia, Anaplasma, tick-borne encephalitis	\$1,406 retail
Total Tox Burden	Mycotoxins (29), heavy metals, environmental chemicals. Co-burden that impairs immune function and recovery.	\$989 retail
Organic Acids Test (OAT)	Neurotransmitter metabolites, mitochondrial markers, fungal overgrowth. Neurological biochemistry.	\$571 retail
Vibrant Hormone Zoomer	Full thyroid panel, TPO/TgAb antibodies, adrenal cortisol, sex hormones. For Lyme-thyroid assessment.	\$710 retail
Vibrant Gut Zoomer Complete	Gut ecosystem, permeability markers, secretory IgA, pathogens. Establishes gut picture before antimicrobials.	\$780 retail

The Gut-Lyme Connection

The gut is not incidental to Lyme disease recovery. It is central to it. The relationship between gut health and Lyme disease operates through several specific mechanisms, and each one affects both the course of the infection and the effectiveness of the treatment protocol. Patients who address Lyme without addressing the gut consistently produce worse outcomes than those whose protocols treat both simultaneously.

How Lyme Compromises the Gut

Borrelia burgdorferi and its co-infections produce systemic immune activation that has specific consequences for gut function. Chronic immune activation driven by tick-borne organisms suppresses secretory IgA production, the gut's primary mucosal immune defense. Secretory IgA normally monitors the gut's microbial inhabitants and prevents pathogenic organisms from establishing colonization. When secretory IgA is depleted, the gut becomes susceptible to pathogenic overgrowth and dysbiosis.

The immune dysregulation of Lyme disease also affects the composition of the gut microbiome directly. Th17-dominant immune polarization, which characterizes both active Lyme and Hashimoto's, promotes the growth of specific bacterial populations that generate more inflammatory LPS burden and further suppresses the commensal organisms that maintain gut barrier integrity. The result is a gut microbiome that, in the presence of active Lyme infection, is progressively less capable of maintaining the gut barrier that protects against systemic antigen exposure.

Antibiotic treatment for Lyme disease compounds this problem. Extended antibiotic courses, which are frequently required for late-disseminated Lyme, produce significant microbiome disruption. The diversity depletion from antibiotics can take twelve to twenty-four months to recover without active microbiome restoration. Patients who complete antibiotic courses without microbiome restoration support emerge with a gut that is both depleted of beneficial organisms and vulnerable to recolonization by pathogenic ones.

How Gut Dysfunction Impairs Lyme Recovery

The bidirectionality of this relationship is clinically critical. A compromised gut barrier generates LPS endotoxemia that sustains the systemic inflammatory burden independently of the Lyme infection itself. Even if antimicrobial treatment effectively reduces the *Borrelia* and co-infection burden, the inflammatory signaling from a leaky gut maintains the same immune activation state that impairs neurological function, drives fatigue, and sustains the Th17 polarization that promotes autoimmune consequences. Patients who have completed Lyme treatment and still feel unwell⁴ frequently have persistent gut dysfunction as a primary driver of their ongoing symptoms.

The liver's capacity to process the microbial debris generated during antimicrobial treatment depends on an adequate supply of glutathione, methylation capacity, and bile acid production. All three of these are impaired

by gut dysfunction. A compromised gut produces less bile acid recycling, depletes glutathione through ongoing oxidative stress, and generates the methyl-depleting inflammatory burden that reduces Phase 2 liver detoxification capacity. The patient undergoing Lyme treatment with a compromised gut is attempting to process a high microbial debris load through a detoxification system that is already overwhelmed.

This is the mechanistic explanation for the severity of Herxheimer reactions in Lyme treatment. The antimicrobial is working. But the drainage and detoxification systems are not capable of handling the load it is generating. Preparing the gut, restoring gut barrier function, and establishing drainage capacity before introducing antimicrobials substantially reduces the severity of Herxheimer reactions and improves both tolerability and outcomes.

The Gut Workup for Lyme Patients

For Lyme patients with significant gut symptoms, the Gut Zoomer Complete at \$780 provides the comprehensive microbial picture needed before antimicrobial treatment begins. The secretory IgA result is particularly informative: severely depleted secretory IgA in the context of Lyme disease confirms the expected gut immune suppression and signals that microbiome restoration support must accompany any antimicrobial protocol. For patients without prominent gut symptoms, the Wheat Zoomer at \$479 is a minimum assessment to determine whether gut permeability is contributing to the ongoing inflammatory picture.

Drainage First

The drainage-first principle is not a theoretical preliminary. In Lyme disease specifically, it is the clinical factor that most consistently determines whether treatment is tolerated and whether it produces lasting results. Lyme treatment without drainage preparation is the primary cause of the severe Herxheimer reactions that force patients to stop their protocols, the treatment fatigue that leads providers to conclude that further antimicrobial intervention is futile, and the incomplete recovery that leaves patients partially improved but chronically unwell.

The biological mechanism is straightforward. When *Borrelia* and co-infections are targeted by antimicrobial agents, the organisms die and release their contents into systemic circulation. Spirochete cell wall components, bacterial DNA, endotoxins, and the inflammatory cytokines generated by the dying organisms create a massive acute burden on the lymphatic system, the liver, and the kidneys. If these systems are not adequately prepared to process this load, it accumulates, circulates, and produces the Herxheimer reaction: a worsening of every symptom the patient has, often dramatically, in the days following each treatment dose.⁵

In patients with chronic Lyme disease, all three drainage systems are typically compromised before treatment even begins. The liver has been processing the inflammatory burden of active infection for months or years. Glutathione is depleted. Methylation capacity is reduced by the ongoing inflammatory burden. The lymphatic system is congested from years of chronic inflammation. The kidneys are managing an elevated toxic metabolite load. Introducing intensive antimicrobial treatment into this environment without first restoring drainage capacity is a predictable setup for treatment failure.

The Drainage Protocol for Lyme

Product	Clinical Role	Dosing
Lymph 2 Matrix (Physica)	Primary lymphatic and ECM drainage for the majority of Lyme cases. ECM congestion from years of chronic inflammatory burden is universal in late-disseminated Lyme. Addresses xenobiotic accumulation alongside lymphatic flow.	1 dropper 2x daily in warm water
Lymph 3 Chronic (Physica)	For patients with symptoms present for more than 3 years, heavy metal burden confirmed on Total Tox Burden, or evidence of deep degeneration. Addresses the deepest layer of lymphatic congestion. Never used simultaneously with Lymph 2.	1 dropper 2x daily in warm water
Methyl-B12 Liposome Spray (Physica)	Methylation support for Phase 2 liver detoxification. The liver's capacity to conjugate and eliminate Herxheimer debris depends on methylation capacity. Sublingual delivery bypasses the gut for patients with impaired absorption.	5 sprays sublingual 2x daily
SpectraLyte (Physica)	Cellular hydration and electrolyte balance. Dehydration at the intracellular level is among the most consistent drainage bottlenecks in chronic illness patients.	Per label direction

For acute Lyme presentations (symptoms present less than 12 weeks), Lymph 1 Acute is the appropriate drainage selection. For most chronic Lyme cases, Lymph 2 Matrix is standard. For patients with more than three years of symptoms or confirmed heavy metal burden, Lymph 3 Chronic is the clinical choice. These formulas are never used simultaneously. The drainage protocol runs for a minimum of two to three weeks before any antimicrobial support is introduced. In complex cases with multiple co-infections, confirmed mold burden, and significant chronicity, three to four weeks of drainage preparation is the appropriate standard.

Critical Timing Note: Two to three weeks of drainage before antimicrobials is not optional. It is the difference between a Herxheimer reaction that is clinically manageable and one that forces the patient to discontinue treatment entirely. The investment of preparation time pays the most immediate returns in clinical practice.

Diet and Environmental Intervention in Lyme

Diet in Lyme disease is not primarily about symptom management. It is about creating the biochemical and immunological environment in which antimicrobial treatment is most effective and recovery is most efficient. Several specific dietary principles are supported by the mechanistic understanding of how Lyme affects metabolic and immune function.

Anti-Inflammatory Foundations

The Th17-dominant immune environment of chronic Lyme disease is sustained by the same dietary factors that drive systemic inflammation in other chronic conditions: seed oils high in linoleic acid, refined sugar and high-glycemic carbohydrates, processed foods containing emulsifiers, and alcohol. Eliminating these inputs reduces the inflammatory signaling that drives cytokine production and maintains the immune polarization characteristic of active Lyme. This is not a cure for Lyme. But it removes a significant dietary driver of the inflammation that worsens every Lyme symptom.

The anti-inflammatory dietary foundation for Lyme parallels the one for Hashimoto's because the immune mechanisms overlap substantially. Remove gluten and refined carbohydrates. Remove seed oils and replace with olive oil, avocado oil, and fat from grass-fed or wild-caught sources. Remove alcohol, which is directly hepatotoxic and impairs the liver's capacity to process the microbial debris generated during treatment. Remove processed foods containing emulsifiers and preservatives that promote gut barrier disruption.

Protein and Micronutrient Adequacy

Antimicrobial treatment for Lyme creates a significantly elevated demand for the detoxification substrates that the liver and kidneys need to process microbial debris. Glutathione, the body's primary intracellular antioxidant and detoxification molecule, is synthesized from cysteine, glycine, and glutamine, all of which are derived from dietary protein. A protein-adequate diet, with emphasis on clean animal proteins providing the full amino acid spectrum, is the dietary foundation for glutathione synthesis during active Lyme treatment.

Magnesium deficiency is near-universal in Lyme patients and has direct clinical consequences. Magnesium is required for over 300 enzymatic reactions, including ATP production, DNA repair, and the enzymatic steps in Phase 2 liver detoxification. It is depleted by chronic stress, by the inflammatory burden of active infection, and by the medications commonly prescribed for Lyme. Prioritizing magnesium-rich foods, including leafy greens, pumpkin seeds, dark chocolate, avocados, and almonds, while minimizing the dietary factors that accelerate magnesium depletion (refined sugar, alcohol, excessive caffeine), supports the metabolic foundation for recovery.

Environmental Considerations

Re-exposure prevention is a component of Lyme management that is frequently overlooked once treatment begins. Patients in tick-endemic areas who continue to have regular outdoor exposure without adequate tick prevention measures can experience repeated re-infection that undermines treatment progress. This is not a reason to restrict outdoor activity. It is a reason to be systematic about protective measures: wearing permethrin-treated clothing in wooded or grassy areas, performing full-body tick checks after outdoor exposure, and showering within two hours of potential tick exposure.

Mold burden is the most important concurrent environmental intervention for Lyme patients. The relationship between mycotoxin exposure and Lyme immune suppression is bidirectional: Lyme impairs the immune response to mold toxins, and mold toxins impair the immune response to Lyme. Patients who have both conditions, which is extremely common in clinical practice, require environmental assessment for water-damaged building exposure alongside their Lyme workup. An ERMI or HERTSMI-2 dust test of the living and working environment is the most reliable confirmation of current mold exposure. Active mold toxin burden that is not identified and addressed will consistently limit Lyme treatment outcomes.

The Lyme–Autoimmune and Neurological Connection

Borrelia burgdorferi does not just cause infection. It creates the immunological conditions in which autoimmune cascades initiate, accelerate, and persist. This is why so many chronic Lyme patients arrive with a cluster of autoimmune diagnoses that preceded, coincided with, or followed their tick-borne illness. The connection is not coincidental. It is mechanistic.

Understanding how *Borrelia* triggers autoimmunity changes the clinical picture in an important way: the autoimmune condition a patient develops is not determined by the organism alone. Lyme creates the terrain. The patient's genetic predispositions, prior immune exposures, and tissue vulnerabilities determine which autoimmune expression emerges from that terrain.

How *Borrelia* Triggers Autoimmunity: Five Mechanisms

The first and most well-documented is molecular mimicry. When *Borrelia* antigens share structural similarity with human tissue proteins, antibodies generated against the organism cross-react with self-tissue. This is not a theoretical risk. It is a confirmed laboratory finding for several *Borrelia* antigens and human tissue targets.

The second mechanism is polyclonal B-cell activation. *Borrelia* drives non-specific, broad-spectrum antibody production. This is why Lyme patients frequently show low-positive ANA, rheumatoid factor, and other autoimmune markers on standard labs even when the specific autoimmune diagnosis criteria are not met. The immune system has been pushed into non-specific antibody overproduction, and some of that production inevitably targets self-tissue.

The third mechanism is bystander activation. Chronic inflammation in tissue infected or affected by *Borrelia* activates self-reactive T cells that would otherwise remain dormant. These T cells do not require direct molecular mimicry to become pathogenic. They simply require the inflammatory environment that chronic Lyme creates.

The fourth mechanism is epitope spreading. The initial immune response against *Borrelia* antigens can widen over time to include self-antigens through a process in which the immune system, already primed and activated, begins recognizing adjacent tissue proteins as targets. This is a progressive mechanism that explains why autoimmune disease in Lyme patients can emerge months to years after the initial infection.

The fifth mechanism is sustained Th17 polarization. *Borrelia* drives a Th17-dominant immune response. Th17 cells produce IL-17 and IL-22, which drive tissue inflammation and breakdown of barrier integrity in the gut and elsewhere. Sustained Th17 dominance is the shared immune architecture of Hashimoto's thyroiditis, multiple sclerosis, rheumatoid arthritis, and lupus. Lyme sustains the same polarization that drives all of these conditions.

Documented Autoimmune Associations

Lyme arthritis is the most RCT-confirmed autoimmune manifestation of *Borrelia* infection. Outer surface protein A (OspA) of *Borrelia burgdorferi* shares amino acid sequence homology with lymphocyte function-associated antigen 1 (LFA-1), which is expressed on synovial T cells. The resulting cross-reactive immune attack on joint tissue produces an autoimmune arthritis pattern that persists even after antibiotic eradication in a subset of patients. This is the condition called “post-infectious autoimmune Lyme arthritis,” and it represents confirmed molecular mimicry at the research level.⁶

The connection to demyelinating disease and multiple sclerosis has a more specific mechanistic basis than is widely appreciated. The *Borrelia burgdorferi* flagellin protein, the 41 kDa antigen that produces the most commonly positive band on the standard Western blot, generates an IgM antibody response that has been specifically demonstrated in the laboratory to cross-react with myelin basic protein.⁷ This is documented molecular mimicry between a *Borrelia* antigen and central nervous system myelin. Lyme neuroborreliosis produces white matter lesions on MRI and oligoclonal bands in cerebrospinal fluid that are radiographically and clinically indistinguishable from multiple sclerosis. In tick-endemic regions, any new MS diagnosis warrants *Borrelia* PCR testing before committing to immunosuppressive MS treatment.

The connection to Hashimoto’s thyroiditis and other autoimmune thyroid conditions is proposed primarily through two pathways: the general Th17 polarization that drives thyroid lymphocytic infiltration, and a possible cross-reactivity between *Borrelia* outer surface proteins and thyroid tissue antigens. This connection is less mechanistically confirmed than Lyme arthritis or the MS-mimicry finding, but the clinical co-occurrence is real. Patients presenting with new or refractory Hashimoto’s in tick-endemic areas, or whose antibodies do not normalize despite comprehensive gut repair and gluten elimination, warrant Tickborne 2.0 testing.

Lupus-like and rheumatoid arthritis-like presentations arise primarily from the polyclonal B-cell activation and Th17 polarization mechanisms. Patients with Lyme frequently show low-positive ANA and dsDNA antibodies, morning stiffness, symmetric joint involvement, and fatigue patterns that satisfy rheumatological diagnostic criteria. The distinction from primary autoimmune disease is critical because the treatment diverges sharply: immunosuppression with methotrexate or biologics in a patient with undiagnosed Lyme suppresses the immune response attempting to contain the infection.

The Neurodegeneration Connection: Lyme and Alzheimer’s Disease

The connection between tick-borne illness and neurodegenerative disease is among the most clinically significant and underrecognized areas in this field. Dr. Dale Bredesen at the Buck Institute for Research on Aging developed the ReCODE (Reversal of Cognitive Decline) protocol, which identified multiple distinct subtypes of Alzheimer’s disease. Subtype 3, which he terms “toxic” Alzheimer’s, is characterized by chronic infectious burden, mycotoxin exposure, and heavy metal accumulation, a profile that overlaps precisely with the chronic Lyme and mold patient population.

The mechanistic basis for this connection is stronger than clinical observation alone. Dr. Judith Miklossy at the University of Lausanne published findings of *Borrelia burgdorferi* spirochetes and DNA detected in postmortem brain tissue of Alzheimer's disease patients, published in the *Journal of Neuroinflammation*.⁸ This is a direct detection finding, not a serology-based association.

At Harvard Medical School, researchers Robert Moir and Rudolph Tanzi published evidence in *Science Translational Medicine* that amyloid beta, long considered the pathological hallmark of Alzheimer's disease, has direct antimicrobial activity including activity against *Borrelia burgdorferi*.⁹ Under this antimicrobial protection hypothesis, amyloid deposition in the brain is not a primary pathological process but rather the brain's innate immune defense response to chronic microbial presence. If this hypothesis is correct, treating the chronic infection may be as important as addressing the amyloid itself in a subset of Alzheimer's patients.

The clinical implication is specific: patients presenting with early cognitive decline, word retrieval difficulty, brain fog with a chronic illness pattern, and a history of tick exposure or living in tick-endemic environments warrant comprehensive infectious and toxic burden assessment before a neurodegenerative diagnosis is accepted as idiopathic. Tickborne 2.0 PCR, Vibrant Mycotoxin Profile, and heavy metal testing through Total Tox Burden represent the minimum workup for any patient presenting with the Bredesen Type 3 cognitive pattern.

The evidence framing is important: Lyme-associated demyelination and the Miklossy/Bredesen neurodegenerative connections are supported by specific mechanistic findings and postmortem detection data. They are not RCT-proven causal relationships. The clinical standard for investigation is not proof of causation. It is mechanistic plausibility in the setting of a treatable condition. The cost of missing active Lyme or mycotoxin burden in a patient on the trajectory toward dementia is permanent. The cost of testing is low.

Your Next Steps with Root Health

Chronic tick-borne illness is one of the most complex and mismanaged conditions in conventional medicine, not because the science is unclear, but because the clinical infrastructure for comprehensive testing and individualized treatment sequencing is largely absent from standard practice. The patients who recover are overwhelmingly those who find clinicians willing to test comprehensively, treat based on what testing reveals, support drainage and gut function as foundational elements of the protocol, and manage the co-infection picture rather than just the *Borrelia*.

This guide gives you the framework to understand your own clinical picture and to have more productive conversations with providers. What it cannot do is replace individualized clinical assessment and management. The specific organisms present in your case, their burden levels, the co-infection combination, your gut status, your drainage capacity, your mold burden, and whether thyroid autoimmunity is a concurrent issue all determine how this framework applies to you specifically.

Start with Comprehensive Testing

The minimum comprehensive tick-borne workup is the Vibrant Tickborne 2.0 PCR at \$1,406, available through rootcauseunlocked.com/order-labs. For patients with fatigue that is disproportionate to their overall illness burden, add the Total Tox Burden panel at \$989 to assess the mycotoxin and heavy metal co-burden. For patients with neurological or psychiatric symptoms, add the Organic Acids Test at \$571. For patients with thyroid symptoms or a history of thyroid disease, add the Hormone Zoomer at \$710. Ordering the complete picture in a single draw is more cost-effective and faster than sequential ordering and produces a complete clinical picture that makes treatment planning precise rather than speculative.

Book a Root Discovery Session

A Root Discovery Session is a complimentary 30-minute clinical conversation with Dr. Seay. It is not a sales call. It is a focused clinical intake where your symptom history, tick exposure history, prior testing, and current clinical picture are reviewed to identify the most targeted path forward for your specific case. For patients who have been through multiple providers without a satisfying answer, this conversation often clarifies the diagnostic picture significantly. Book at rootcauseunlocked.com/discovery.

The Root Restore Program

For patients ready for structured, supervised 1-on-1 clinical care, the Root Restore Program is a 6-month protocol that includes comprehensive testing, a personalized drainage-first supplement and antimicrobial protocol, dietary guidance, and ongoing clinical oversight throughout every phase. Payment plans are available. Learn more at rootcauseunlocked.com/lyme.

A Note from Dr. Seay

If you have been told your Lyme test is negative but your symptom history is consistent with tick-borne illness, a negative test is not the end of the investigation. It is a reason to use a better test.

If you have been treated for Lyme but have not fully recovered, incomplete pathogen identification or unaddressed co-infections are the most consistent clinical explanations. The answer is usually in the data you do not yet have.

Dr. Adam Seay, DC · Root Health, Garner NC · rootcauseunlocked.com

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From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek medical care now for facial droop, severe headache with neck stiffness, palpitations or fainting, or new severe joint swelling. Lyme can affect the nerves, heart, and joints, and those presentations need prompt treatment rather than a testing strategy.

This is the most consequential guide in this kit, because Lyme is a real bacterial infection where **delay changes outcomes** and where a large market of unproven treatment sits alongside the real one.

After treatment, when symptoms persist

This section was missing, and for a meaningful share of readers it is the section they need most.

The situation has a name and it is recognized. Post-treatment symptoms following Lyme disease sit within a broader group of infection-associated chronic conditions that also includes post-COVID conditions and myalgic encephalomyelitis. A 2026 review in the infectious disease literature argues for recognizing these conditions, notes that millions of Americans are affected, and calls for better surveillance, research, and inclusive trial design (PMID 41967005).

That matters because being told your symptoms are unexplained, after documented treatment for a documented infection, is its own injury. The condition is real, it is increasingly acknowledged, and the honest state of things is that treatment is not settled.

Pacing is the practical centerpiece. A scoping review of pacing in ME/CFS describes it as a prominent energy-management strategy while stating plainly that the literature is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS puts energy management at the center of daily life, alongside sleep, diet, and psychological support (PMID 39961545).

Naming the population honestly: that research is in ME/CFS rather than in post-treatment Lyme. It is included because the clinical problem overlaps and because the safety principle is the same one used everywhere on this shelf: **if your symptoms reliably worsen a day or two after exertion, graded pushing is the wrong instruction.**

What that looks like in practice.

- Find your envelope on an average day, not a good one, and plan to the average.
- Break activity into shorter blocks with rest between, rather than one long push.
- Treat sleep as a treatment. Fixed window, seven to nine hours.
- Five to ten minutes daily of slow breathing with a longer exhale, for the nervous system load that chronic illness carries.
- Expect a non-linear course, and judge progress in months rather than days.

And the thing that has to be said plainly. Persistent symptoms after treatment are not evidence that you need indefinite antibiotics. Long-term antibiotic therapy for persistent symptoms has real harms, and that decision belongs with an infectious disease physician rather than with a protocol.

If you were bitten in the last few days, read this section first

There is a specific, evidence-backed action, and it has a time window.

A systematic review and meta-analysis of antibiotic prophylaxis after tick bite pooled 6 studies covering 3,766 people. Unfavorable events occurred in **0.4% of those treated versus 2.2% of controls**, a pooled risk ratio of **0.38**.¹

The subgroup analysis is the useful part. **Single-dose 200 mg doxycycline had the strongest result, at a risk ratio of 0.29**. A 10-day antibiotic course was not statistically significant, and topical azithromycin was not either. The authors concluded the evidence supports antibiotic prophylaxis and shows an advantage for the single dose.

So: after a qualifying tick bite, a single dose of doxycycline is the intervention with the best evidence, and it is time-limited. That is a same-week conversation with a clinician, not something to research for a month. Whether you qualify depends on tick species, attachment time, and local infection rates, which is exactly what the visit is for.

Do not wait for a test to decide. Antibodies take weeks to develop. Testing in the days after a bite tells you about past exposure, not about this bite.

The single most important testing principle

A rash that looks like erythema migrans is a clinical diagnosis. It does not require a positive blood test, and treatment should not wait for one.

This is where the most avoidable harm happens. Someone presents early, the serology is negative because antibodies have not developed yet, and treatment is deferred. The negative result was not evidence of no infection; it was evidence that not enough time had passed.

Why: serologic tests detect your immune response, not the organism. Early in infection that response has not yet formed. **A negative test early in the illness does not exclude Lyme disease.**

That principle is why the guideline-based approach treats a characteristic rash in an endemic area on clinical grounds.

What the standard testing actually is

Two-tier serology has been the standard: a first-tier immunoassay, then confirmation. **Modified two-tier testing**, using two immunoassays rather than an immunoblot, is now in use and has been compared directly against the standard algorithm.² Single-tier approaches are under development.³

What the tests do well: later-stage disease, where the antibody response is established.

What they do poorly: the first weeks, for the reason above.

What they cannot do at all: tell you whether an infection has been cured, or diagnose persistent symptoms after treatment. More on that below.

The authoritative reference is the 2021 clinical practice guideline developed jointly by the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology.^{4,5} If you want one document behind your care, that is it.

Persistent symptoms after treatment, taken seriously and described accurately

Most people recover fully. The prognosis after appropriate antibiotic treatment of early or late Lyme disease is favorable.⁶

Some do not, and that is a recognized phenomenon. Post-treatment Lyme disease syndrome describes fatigue, musculoskeletal pain, and cognitive complaints persisting **six months or longer** after completing antibiotic therapy.⁶ A 2022 review reaches the same position: most patients fully recover with recommended therapy, while some report persisting nonspecific symptoms.⁷

Known risk factors include delayed diagnosis, greater symptom severity, and neurologic symptoms at the time of initial treatment.⁶ The first of those is the argument for not deferring treatment while waiting on a test.

And this is the part that matters for your labs: two-tier serologic testing is **neither sensitive nor specific for diagnosing PTLDS**, because antibody responses vary a great deal after treatment of early disease.⁶

Read that carefully in both directions. A positive antibody test months later does not prove ongoing infection, because antibodies persist after successful treatment. A negative one does not prove your symptoms are not real. **Serology cannot answer the question you most want answered**, and any test being sold as though it can is overreaching.

Optimal treatment of PTLDS awaits better understanding of the underlying mechanism.⁶ That is an unsatisfying sentence and it is the accurate one.

A caution about specialty testing and treatment

There is a documented market here. A study in Clinical Infectious Diseases set out specifically to identify and characterize the range of **unorthodox alternative therapies marketed to treat Lyme disease**, noting that patients with medically unexplained symptoms or alternative diagnoses who believe they have chronic Lyme are commonly targeted by providers of these therapies.⁸

I am not telling you your symptoms are imagined. I am telling you that a market exists which is built on the gap between real persistent symptoms and the absence of a proven treatment, and that being in that gap makes people vulnerable.

Practical filters for any test being offered:

- Is it performed by a lab accredited for this purpose?
- Has the assay been validated against a reference standard, or does its interpretive framework exist only in the marketing?
- If the result is positive, what specifically changes about my care?
- Does the person offering the test also sell the treatment?

And one thing to do regardless. Tell your physician everything you are taking. Complementary and alternative medicine use is common, reported by 36.7% of American adults in 2022, and **most patients do not disclose it to their physicians**, who may then underestimate the biological potency of what is being taken.⁹ With Lyme in particular, where people often run herbal antimicrobial protocols alongside prescription treatment, non-disclosure is a genuine safety issue rather than a courtesy problem.

Reading your results as a pattern

Characteristic rash, any test result. Clinical diagnosis. Treat.

Recent bite, no symptoms, negative test. Expected. The test is measuring the wrong thing this early. The prophylaxis conversation is the useful one.

Positive two-tier, compatible symptoms, exposure history. Treat per guideline.

Positive antibodies months after treated Lyme, ongoing symptoms. Antibodies persist after cure. This does not establish ongoing infection, and PTLDS is not diagnosed serologically.⁶

Negative standard testing, positive specialty-lab result, no exposure history. Apply the filters above before building a treatment plan on it.

Persistent symptoms, everything negative. Real symptoms, unproven cause. Worth pursuing the rule-outs that produce the same picture, including thyroid disease, anemia, sleep apnea, and vitamin deficiencies.

What to ask your clinician

- “I was bitten recently. Do I qualify for single-dose doxycycline prophylaxis?”
- “I have a rash. Does it need a positive test before treatment?”
- “My symptoms started weeks ago and the test was negative. Should it be repeated now that time has passed?”
- “I finished treatment and still feel unwell. What does the evidence say about what happens next?”
- “Here is everything I am taking, including supplements and herbs.” Bring the actual labels.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, or new severe joint swelling.

This week: any recent tick bite, because prophylaxis is time-limited.

Promptly: an expanding rash after possible exposure.

Track what changes

Root Tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab

Free, works anywhere. Where symptoms persist and testing cannot settle the question, a symptom record over months is genuinely one of the more useful things you can bring to an appointment.

Working with Root Health

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utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab)

Consultations are available to people located in North Carolina, where I am licensed. **If you have an active or suspected acute infection, that belongs with a physician promptly rather than with me.** I will say so.

The keys this kit follows

Unlock Your Lyme works through a set of root causes rather than a single organism, and this kit follows the same order: why the standard test fails, the co-infection picture, the right testing framework, the gut-Lyme connection, drainage first, diet and environment, and the autoimmune and neurological tail.

The reason that structure matters here more than in most conditions: Lyme is the condition where people most often chase the organism for years while the drivers that keep them unwell, sleep debt, an unrepaired gut, a nervous system stuck in threat, and an activity pattern that guarantees crashes, go unaddressed. Killing something is not the same as recovering.

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing here replaces antibiotic treatment. Lyme is a bacterial infection and delaying treatment to try a dietary approach is the most consequential mistake available in this condition, because delayed diagnosis is a documented risk factor for persistent symptoms afterward. The lab guide in this kit covers that, along with the time-limited action after a tick bite.

Why this guide is not called “The Lyme Diet.” I looked for evidence that a particular diet affects Lyme outcomes and did not find it. What I did find is a real, specific, well-evidenced dietary job that Lyme patients face more than almost any other outpatient group: **recovering from the antibiotics.**

The thing that makes Lyme different

Lyme patients take antibiotics. Often for weeks, sometimes in repeated or extended courses. That is appropriate treatment and it also has a measurable cost that gets very little attention.

Antibiotics reshape the gut microbiome, and recovery takes months rather than days. In a study using shotgun metagenomics, 12 healthy men were followed over **six months** after a four-day antibiotic intervention. The initial changes included blooms of enterobacteria and other pathobionts including *Enterococcus faecalis* and *Fusobacterium nucleatum*, alongside **depletion of Bifidobacterium species and butyrate producers**.¹

Four days of antibiotics. Six months of follow-up. That is the scale of what a multi-week Lyme course is doing.

Two things follow.

The gut symptoms that appear during or after Lyme treatment are frequently the antibiotics, not the infection and not a new mystery illness. That reframing alone spares people a lot of worry and a lot of unnecessary testing.

And the useful dietary work is largely about protecting the microbiome during treatment and rebuilding it afterward. That is concrete, evidenced, and time-bound, which is more than most condition diets can say.

During treatment

Probiotics, which have the strongest number in this guide

Two independent meta-analyses agree closely. One pooled 42 studies with 11,305 participants and found co-administration of probiotics with antibiotics **reduced the risk of antibiotic-associated diarrhoea in adults by 37%**, a risk ratio of 0.63 with a confidence interval of 0.54 to 0.73.² Another pooled 36 studies with 9,312 participants and found a **38% reduction**, a pooled relative risk of 0.62 with a confidence interval of 0.51 to 0.74, with the protective effect holding across reasons for antibiotic treatment, duration, and dose.³

Two large analyses, different study sets, effectively the same answer. That is about as good as supplement evidence gets.

Practically: start when the antibiotics start rather than after problems appear, separate the probiotic from the antibiotic dose by a couple of hours, use named strains at stated counts, and continue for a period after the course finishes.

One caution worth stating: if you are immunocompromised, live probiotic organisms are a genuine clinical question. Ask first.

Food during the course

Eat, even when appetite drops. Under-eating during treatment slows everything downstream.

Keep fiber in. It feeds the butyrate producers that the antibiotics are depleting.¹ If bloating makes that hard, go with cooked vegetables and soluble sources rather than a raw salad.

Fermented foods if tolerated. Yogurt, kefir, sauerkraut, kimchi.

Take doxycycline with a full glass of water and stay upright afterward, because it can irritate the oesophagus, and be aware that calcium, iron, and magnesium reduce its absorption. Separate mineral supplements and large dairy servings from the dose. Doxycycline also causes photosensitivity, which is a sunburn risk rather than a diet issue but the most common surprise of the course.

Alcohol off during treatment. It burdens the same organ clearing the medication and does you no favours.

After treatment: rebuilding

The rebuild is where diet does its real work, and the timeline from the microbiome study is the honest expectation: this is months, not a fortnight.¹

Fiber, increased gradually and from varied sources. Variety matters more than volume. Vegetables, legumes, whole grains, fruit, nuts, seeds. Increase slowly, because a fast jump after antibiotics produces bloating that gets blamed on the food.

Fermented foods regularly, not occasionally.

Protein sufficient for recovery, at every meal.

Polyphenol-rich foods. Berries, olive oil, herbs, tea, cocoa.

Rebuild the plate rather than the pill count. The microbiome study shows depletion across whole functional groups, and no capsule replaces a varied diet.

What I would not do: start a restrictive elimination diet during or immediately after treatment. Your microbiome is already depleted and restriction narrows it further at exactly the wrong moment.

For persistent symptoms

If fatigue, pain, and cognitive complaints continue six months or more after adequate treatment, the lab guide covers what that is and what testing can and cannot say about it.

From a food standpoint, the useful targets are the ones that mimic it. Iron, B12, and vitamin D deficiency each produce fatigue and cognitive fog and each is testable and correctable. Persistent symptoms after Lyme get attributed to the infection when a deficiency is sitting untreated underneath, and that is a genuinely common miss.

Blood sugar stability matters for the day-to-day pattern. Protein at every meal, no liquid sugar, and a short walk after the largest meal do more for afternoon crashes than any protocol.

Eating for sleep, because sleep loss lowers pain thresholds and this population is often in a pain-and-poor-sleep loop. Caffeine earlier, alcohol not used as a sleep aid, and the last large meal away from bedtime.

A Mediterranean-style pattern is a reasonable default. It is not a Lyme treatment, and I am not going to present it as one.

Running a fair experiment

Do the rebuild first, before you experiment with restriction. Give it eight to twelve weeks.

Then one change at a time, with a written baseline, because gradual change is invisible from the inside.

Give restrictions an endpoint. If eight weeks produced nothing, stop. Permanent restriction without benefit is the most common self-inflicted harm in this space, and stacking eliminations on a depleted microbiome makes it worse.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-diet

Realistic expectations

No diet treats *Borrelia*. Antibiotics do, and most people fully recover with recommended therapy.

Probiotics alongside antibiotics reduce antibiotic-associated diarrhoea by roughly 37 to 38% across two large meta-analyses.^{2,3} That is the strongest claim in this guide and it is about tolerating treatment, not curing infection.

Microbiome recovery after antibiotics runs on a months-long clock.¹ Expecting to feel rebuilt in two weeks sets you up to abandon something reasonable.

And where symptoms persist, the highest-value dietary work is correcting the deficiencies that produce the same picture, which is testing followed by food rather than a protocol.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, new severe joint swelling. Also severe or bloody diarrhoea during or after antibiotics, which can indicate *C. difficile* and is a medical problem rather than a dietary one.

Promptly: inability to keep food or fluids down during treatment.

Before a restrictive diet: if you are underweight, have a history of disordered eating, are pregnant, or are managing diabetes.

Working with Root Health

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utm_source=guide&utm_medium=ebook&utm_campaign=lyme-diet](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-diet)

Consultations are available to people located in North Carolina, where I am licensed. **If you have an acute or suspected active infection, that belongs with a physician promptly**, and I will say so rather than take the appointment.

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The nervous system and the stress axis

This belongs in a Lyme kit for a specific reason: sustained illness, contested diagnosis, and months of uncertainty are a genuine physiological load, not a soft one.

What I will claim and what I will not. There is no trial showing that stress reduction clears an infection, and I am not going to imply one. What the literature on infection-associated chronic conditions and on fatigue-dominant illness does consistently include is psychological support as a component of comprehensive care alongside sleep, diet, and energy management (PMID 39961545, PMID 41967005).

Why it matters practically. A nervous system held in threat degrades sleep, and poor sleep worsens pain, cognition, and fatigue. That loop is enterable from either end, and the end you can reach without a prescription is the daily one.

The work:

- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
- A fixed sleep window, protected like a medication.
- Pacing, covered in the lab guide, which is the single most important behavioral instruction in this condition.
- Eat sitting down and unhurried, because digestion is parasympathetic and the gut work in this kit depends on it.

And the thing worth saying out loud. Being disbelieved is part of this illness for many people, and it is exhausting in its own right. That is a reason to bring someone competent alongside you, not a reason to conclude the symptoms are imaginary.

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Doxycycline administration and interaction guidance reflects the product prescribing information.

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Seek medical care now for facial droop, severe headache with neck stiffness, palpitations, fainting, or new severe joint swelling.

Read the lab guide first. It covers the single time-limited action that matters most, prophylaxis after a tick bite, and why a negative early test does not exclude infection.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

The one instruction in this guide that is not optional

Tell your physician everything you are taking.

Complementary and alternative medicine use was reported by **36.7% of American adults in 2022, and most patients do not disclose it to their physicians**, who may consequently underestimate the biological potency of what is being taken.¹

In Lyme this is not a courtesy issue, it is a safety issue. People frequently run herbal antimicrobial protocols alongside prescription antibiotics. Herbs are pharmacologically active, some affect the same liver enzymes that clear medications, and some affect bleeding. Your prescriber cannot account for what they do not know about.

Bring the actual labels, not product names, because blends contain things the name does not mention.

Where the honest boundary sits

Lyme disease is a bacterial infection. Its treatment is antibiotic, guided by a joint guideline from the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology.² **Most patients fully recover with recommended therapy.**³

No supplement treats the infection, and delaying antibiotic treatment to try one is the single most consequential mistake available in this condition, because delayed diagnosis is a documented risk factor for persistent symptoms afterward.⁴

So what is a supplement guide for here? Two legitimate things: supporting general recovery, and helping you evaluate the very large market you are about to encounter. The second is worth more.

The market, described accurately

A study published in Clinical Infectious Diseases set out specifically to identify and characterize the range of **unorthodox alternative therapies marketed to treat Lyme disease**. Its framing is worth quoting in substance: patients with medically unexplained symptoms or alternative diagnoses who believe they chronically have Lyme are **commonly targeted by providers of alternative therapies**.⁵

Here is why that market exists, and I want to be fair to everyone in it. **Post-treatment Lyme disease syndrome is real**. Fatigue, musculoskeletal pain, and cognitive complaints persisting six months or longer after adequate antibiotic therapy is a described phenomenon with recognized risk factors.⁴ And there is no proven treatment for it; optimal management awaits better understanding of the mechanism.⁴

A real condition with no proven treatment is exactly the space where an industry grows. That does not make everyone in it dishonest, and it does mean the burden of proof is on the seller.

Filters worth applying to anything offered to you:

- Does the person recommending it also sell it?
- Is the claimed mechanism testable, or does worsening also count as evidence it is working?
- Is there an endpoint, or is this indefinite?
- What would have to happen for them to tell you to stop?

That third one catches most of it. A protocol with no defined endpoint is a subscription.

On prolonged antibiotic therapy specifically. Extended antibiotic courses for post-treatment symptoms have been the subject of controlled trials, and the literature on them is limited and has been reviewed.⁶ This is a decision for a physician working from the guideline,² not something to pursue through a supplement protocol or an out-of-network prescription without that framing. Long-term intravenous antibiotics carry real risks including line infections.

What is reasonable

The unglamorous foundation, which is most of what is defensible

Sleep, protein sufficient for recovery, and correcting documented deficiencies in iron, B12, and vitamin D. The fatigue and cognitive symptoms attributed to persistent Lyme overlap almost exactly with those deficiencies, with thyroid disease, and with sleep apnea. Those are testable and treatable, and they are in the rule-out list in the lab guide.

That is not an exciting recommendation. It is the one with evidence behind it.

Pacing, if post-exertional crashes are part of your picture

Where activity reliably produces a delayed crash, pace by what you can repeat rather than by what you can manage on a good day. This costs nothing and prevents the boom-and-bust pattern that turns a slow recovery into a flat one.

Probiotics during and after antibiotics

Reasonable, low risk, and worth using for the antibiotic-associated gut effects rather than as anything anti-Borrelia. Named strains, and separate from the antibiotic dose by a couple of hours.

Ask your practitioner first

No products and no links here.

Any herbal antimicrobial protocol, if you are on prescription treatment. This is the interaction question above, and it is the most likely place for real harm. *Ask:* “Here are the labels of everything I am taking. Does any of it interact with my treatment or affect my liver enzymes?”

Anything at all, if you take an anticoagulant. Several popular botanicals affect bleeding.

Prolonged or intravenous antibiotics. A physician decision, made against the guideline, with the risks stated.^{2,6}

Any therapy involving a device, an infusion, or a procedure. Those are the highest-cost and highest-risk end of the market described above,⁵ and they deserve the filters.

What I deliberately left out, and why

The herbal antimicrobial protocols. They are the centrepiece of the alternative Lyme market. I searched for clinical trial evidence in patients and did not find it. Laboratory activity against *Borrelia* is not the same as a human outcome.

Anything sold as eradicating persistent infection. Whether persistent infection causes PTLDS is an open scientific question, not a settled premise you can buy a solution to.

Protocols with die-off or Herxheimer framing. A model in which feeling worse confirms the treatment is working cannot be tested against reality, and unfalsifiable is a warning sign rather than a feature.

Things I use in practice that are not in here. A guide sold to someone I have never examined has no history, no medication list, and no way to change course. Where the evidence does not support recommending something at a distance, it stays out. In this condition more than any other in the kit, I would rather send you to a physician than sell you a plan.

Where to find quality versions of these

Everything above was written without reference to any product line, and this is the guide with the least to buy.

Third-party testing first. NSF Certified for Sport and USP Verified.

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of sales through that link. Nothing here was included, excluded, or ranked based on what it earns, and the largest-selling category in the Lyme space is the one this guide declines to recommend.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, new severe joint swelling.

This week: any recent tick bite. Prophylaxis is time-limited and covered in the lab guide.

Before starting anything: if you are on prescription treatment, an anticoagulant, or have liver disease.

Working with Root Health

Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?
utm_source=guide&utm_medium=ebook&utm_campaign=lyme-supplement](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-supplement)

Consultations are available to people located in North Carolina, where I am licensed. **If you have an acute or suspected active infection, that belongs with a physician promptly.** I will tell you so rather than take the appointment.

If symptoms persist after treatment

Read the full section in **Understanding Your Lyme Labs**, because it is the part of this condition where the wrong instruction does the most harm. The short version:

Post-treatment symptoms sit within the recognized group of infection-associated chronic conditions (PMID 41967005). The practical centerpiece is pacing rather than pushing. A scoping review of pacing in myalgic encephalomyelitis and chronic fatigue syndrome describes it as a prominent energy-management strategy while stating that the literature is not yet sufficient to dictate practice (PMID 37838675), and co-production work with adults living with ME/CFS puts energy management at the center of daily life alongside sleep, diet, and psychological support (PMID 39961545).

That research is in ME/CFS, not post-treatment Lyme, and I would rather name the population than imply a match. The reason it applies is the shared safety principle used across this shelf: if symptoms reliably worsen a day or two after exertion, graded pushing is the wrong instruction.

No supplement in this guide treats that. Sleep, pacing, and a daily down-regulating practice do more for it than anything you can buy, and persistent symptoms are not a reason for indefinite antibiotics, which is an infectious disease physician's decision.

The keys this kit follows

Unlock Your Lyme works through a set of root causes rather than a single organism, and this kit follows the same order: why the standard test fails, the co-infection picture, the right testing framework, the gut-Lyme connection, drainage first, diet and environment, and the autoimmune and neurological tail.

The reason that structure matters here more than in most conditions: Lyme is the condition where people most often chase the organism for years while the drivers that keep them unwell, sleep debt, an unrepaired gut, a nervous system stuck in threat, and an activity pattern that guarantees crashes, go unaddressed. Killing something is not the same as recovering.

The nervous system and the stress axis

This belongs in a Lyme kit for a specific reason: sustained illness, contested diagnosis, and months of uncertainty are a genuine physiological load, not a soft one.

What I will claim and what I will not. There is no trial showing that stress reduction clears an infection, and I am not going to imply one. What the literature on infection-associated chronic conditions and on fatigue-dominant illness does consistently include is psychological support as a component of comprehensive care alongside sleep, diet, and energy management (PMID 39961545, PMID 41967005).

Why it matters practically. A nervous system held in threat degrades sleep, and poor sleep worsens pain, cognition, and fatigue. That loop is enterable from either end, and the end you can reach without a prescription is the daily one.

The work:

- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
- A fixed sleep window, protected like a medication.
- Pacing, covered in the lab guide, which is the single most important behavioral instruction in this condition.
- Eat sitting down and unhurried, because digestion is parasympathetic and the gut work in this kit depends on it.

And the thing worth saying out loud. Being disbelieved is part of this illness for many people, and it is exhausting in its own right. That is a reason to bring someone competent alongside you, not a reason to conclude the symptoms are imaginary.

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Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com